CSE350: Digital Electronics & Pulse Techniques

Lecture 08: TTL Fanout, Output Stages, TTL with Totem Pole

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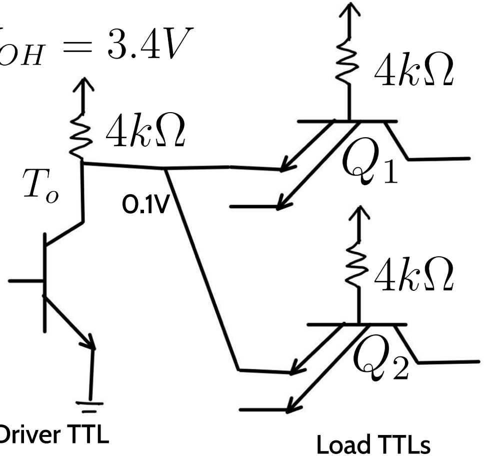
Ex-1: *Find the maximum fanout of the TTL NAND gate.* $\beta_F = 25 \;\; \beta_R = 0.1 \;\; V_{OH} = 3.4V$
$V_{CE} = 0.1V$



Here, we have only showed the parts of the NAND gates that we need for our fanout calculation. The rest parts' calculations are similar to the previous example shown in *Lec-07*. We will use those results directly here to keep things simple, but in exam you must show the full calculation.

Case-Output of driver low: This will result in either (0,1) or (0,0) at the loads' input. Either way, the other two transistors except the input ones will be in cutoff for all loads. We have seen that in these two cases, current flows out from the emitters of the multi-emitter input transistors. It is maximum for (0,1), where the full base current comes out through the low input emitter. It was found to be 1.025mA.

$$I_{B_1} = \frac{5-0.9}{4k} = 1.025mA$$

Now, the no load collector current of the output transistor of driver circuit is, $I_c' = \dfrac{5-0.1}{4k} = 1.225mA$

Also, we found, $I_{E_{1x}} = I_{B_1} = 1.025mA \;\; I_{B_o} = 2.5725mA$

So, considering the marginal condition between saturation and forward active mode, maximum possible collector current of the driver, $I_{C_o,max} = \beta_F * I_{B_o} = I_c' + N * I_{E_{1x}}$

$$\beta_F * I_{B_o} = 25 * 2.5725 = 64.313 = 1.225 + N * 1.025$$

$$N = \lfloor 61.35 \rfloor = 61$$

Thus, maximum fanout for this case is 61. 0.25kΩ

Case-Output of driver high:

The output of the driver will be $V_{OH} = 3.4V$

The loads' input multi-emitter transistors will be in reverse active mode and draw current from the driver this time. We found from Lec-07 that, this amount of current was, $I_{E_{1x}} = I_{E_{1y}} = I_{B_1} * \beta_R = 0.0675mA$

The current from the 5V source of the output transistor of the driver gate will thus flow to the loads' emitters.

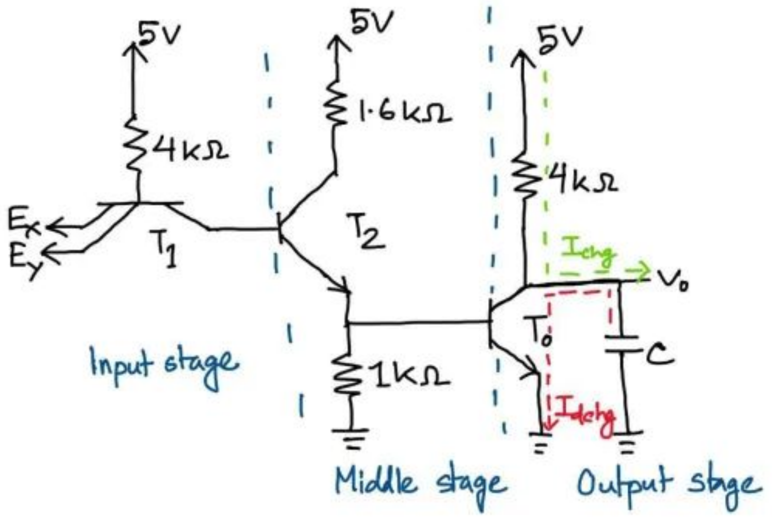
Maximum suppliable current = $\dfrac{5 - 3.4}{4k} = 0.4mA$

The overall maximum fanout
= min(5, 61) = 5

Thus, maximum fanout of this case = $\left\lfloor \dfrac{0.4}{0.0675} \right\rfloor = \lfloor 5.92 \rfloor = 5$

Output Stage Classification of TTL

The output of a TTL circuit can be connected to other circuits. These other circuits' effect can be modeled as a fictitious capacitor. The TTL circuit will then be something like this:



When the output of the TTL is high, the capacitor is charged to a high voltage, while in case of low output, the capacitor is discharged. Thus during the transition from high to low, there needs to be discharge of the capacitor & in the opposite transition, a charging must take place. The charging & discharging currents involved in these two cases are shown in green & red color in the above figure respectively.

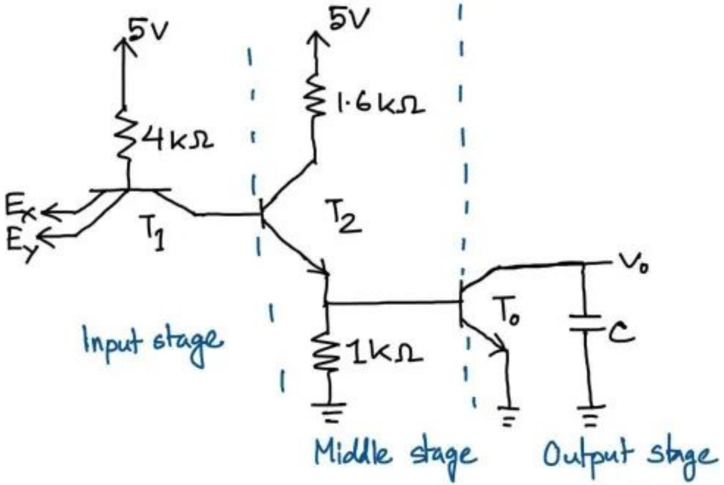
Consequently, the 5V source alongwith the collector resistance of T_o pulls up the the capacitor voltage, and is called the Pull-up Network. On the other hand, The rest of T_o is involved in bringing the voltage down. So, it is called the Pull-down Network.

The TTL circuit can be divided into the 3 stages shown in the previous figure. The configuration of the output stage can be modified and we get the following 3 types of TTLs based on the output stages:

1. Open Collector Output Stage
2. Totem-Pole Output Stage
3. Tristate Output Stage

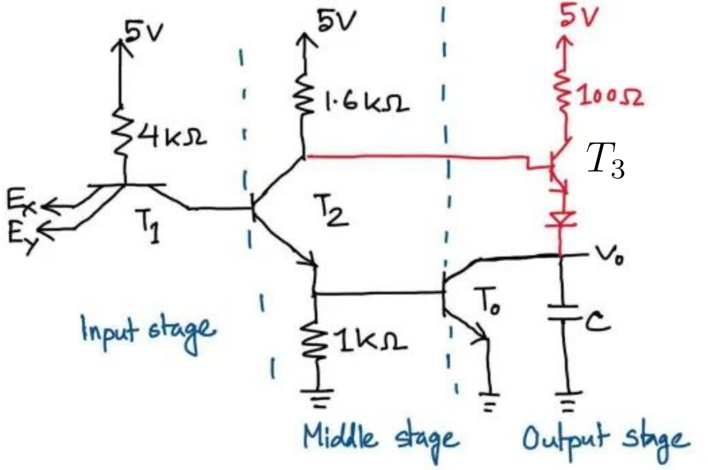
Open Collector Output Stage

The collector terminal of the output stage is an open circuit. Usge- Wired AND gate, open port.



Totem-Pole Output Stage

The part marked in red is the modification made from the base TTL circuit. This output stage acts faster than the open collector type, since it's made of active element like BJT. BJT has a lower resistance, so charging current is higher & takes less time to charge, while not significantly increasing the power consumption.

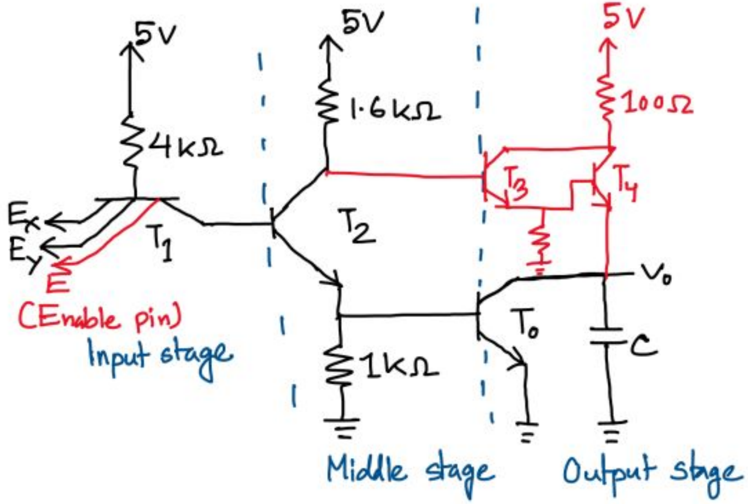


Tristate Output Stage

There are three states of output of this configuration-

1. High
2. Low
3. High Impedance (open circuit)

If the enable pin is off, then all transistors except T_1 will be cutoff.



Then the output terminal will be disconnected from rest of the circuit and will be floating like an open circuit. Since in open circuit we have a high impedance, this state is called high impedance state.

Totem-Pole NAND Gate

Case-1: At least one input is low: T_1 is in saturation, T_2 & T_o will be cutoff. The state of T_3 can be determined from the following analysis:

The base voltage of T_3 will be lower than the collector voltage, because there is higher resistance at base. Thus, $V_{BC} < 0$

Now for low input, output must be high, it must thus be in forward active mode rather than cutoff. However, in forward active mode, there will be 0.7V drop between base and emitter. Also for diode, there will be another 0.7V drop. Thus, the capacitor (modeled as a representation of the connected loads) can be charged to at best a value of 5 - 0.7 - 0.7 = 3.6V. Thus we can conclude that the high output voltage of Totem-Pole circuit will be 3.6V.

Case-2: All input voltages are high: T_1 is in R.A. T_2 & T_o will be in saturation.

There will be 0.8V between base-emitter of the saturated transistors & 0.7V between base-collector (since we are considering 0.1V between collector & emitter). Also for reverse active mode, we know that the base voltage is greater than collector voltage by 0.7V. Thus we have the following voltages at terminals:

Now, our output should be 0.1V for this case. The diode of the output stage will need 0.6V to turn on & T_3 will also need 0.5V. Thus, we need a total of (0.1+0.6+0.5) = 1.2V at the base. But as we can see, we are only getting 0.9V. So, T_3 and the diode will be off. The current I_3 will be 0, and all other analysis are similar to open collector output stage.